

Entangling Capacity of a Local Adaptive Fermion Circuit

Class-A Fermionic Adaptive-Circuit Project

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We explain how locality constrains entanglement production in a fermionic circuit with measurements and outcome-dependent feedback. Starting from a single conditional trajectory, we derive a bound on Schmidt-rank growth and then convert it into a physical-depth bound. We specialize the argument to occupation measurements of finite-range overcomplete Wannier modes followed by fresh-ancilla fermionic-swap correction, deriving the conditional operators and their cut-crossing ranks explicitly. For a strip intersecting two critical interfaces in a two-dimensional system, product-state preparation and conformal entropy scaling imply the necessary resource $D_{\text{circ}} N_x = \Omega(\log N_y)$. This is a restriction on preparation, not a proof of criticality or a prediction for convergence time.

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I. CONDITIONAL DYNAMICS AND ENTANGLEMENT ACROSS A CUT

Measurements and classical feedback can prepare correlations in ways that a local unitary circuit cannot. Nevertheless, a bounded quantum operation does not acquire unlimited capacity to create entanglement across a spatial cut. The quantity appropriate to a bound valid for each measurement branch is the logarithm of Schmidt rank. We develop this argument following Appendix D of Lu, Lessa, Kim, and Hsieh [1], then specialize it to the fermionic controller motivated by Bhuiyan, Pan, and Jian (BPJ) [2].

For a pure input, a measurement with outcome m is specified by a Kraus operator $\hat{K}_m$ and a nonnegative weight w_m . The positive-operator-valued measure (POVM) condition and Born probability are

$$\sum_m w_m \hat{K}_m^\dagger \hat{K}_m = \hat{1}, \quad p_m = w_m \langle \psi | \hat{K}_m^\dagger \hat{K}_m | \psi \rangle. \quad (1)$$

We subsequently absorb $\sqrt{w_m}$ into the branch operator; this scalar does not change its rank. Outcome-dependent unitary feedback is also included in that operator. For a full record $\mathbf{m} = (m_1, \dots, m_M)$, the many-body evolution operator (mEO) and normalized branch are

$$\hat{V}_{\mathbf{m}} = \hat{O}_{m_M}(\mathbf{m}_{<M}) \cdots \hat{O}_{m_2}(m_1) \hat{O}_{m_1}, \quad (2)$$

$$p_{\mathbf{m}} = \|\hat{V}_{\mathbf{m}} |\psi_{\text{in}}\rangle\|^2, \quad |\psi_{\mathbf{m}}\rangle = \frac{\hat{V}_{\mathbf{m}} |\psi_{\text{in}}\rangle}{\sqrt{p_{\mathbf{m}}}}, \quad p_{\mathbf{m}} > 0. \quad (3)$$

The rightmost factor acts first. Earlier outcomes may determine both the kind and location of a later operation. Fixing a record is a mathematical conditioning, not a prescription to postselect it experimentally.

Choose a spatial bipartition $A \cup \bar{A}$ and a fermionic mode ordering adapted to it. Here A denotes a region, not the ancillary system; ancillary quantities carry the subscript a . A graded Fock-space factorization can be represented by an ordinary tensor product with explicit parity strings. Physical branch operators are taken to have definite fermion parity, as do all operators in the controller below; physical pure input states have definite total parity. In that representation, the Schmidt decomposition is

$$|\psi\rangle = \sum_{i=1}^{\chi_A} \sqrt{p_i} |\phi_i\rangle_A \otimes |\theta_i\rangle_{\bar{A}}, \quad p_i > 0, \quad \sum_i p_i = 1. \quad (4)$$

Both vector sets are orthonormal, and the reduced density operator has exactly χ_A nonzero eigenvalues. With natural logarithms,

$$S_0(A) = \log \chi_A, \quad S_q(A) = \frac{\log \sum_i p_i^q}{1-q}, \quad S_1(A) = -\sum_i p_i \log p_i. \quad (5)$$

The Hartley entropy S_0 counts components regardless of their weights. The von Neumann entropy is the continuous $q \rightarrow 1$ limit; for all $q > 0$, $S_q \leq S_0$.

A. Why von Neumann entropy growth is different

A local filter cannot create independent Schmidt components, but it can change their relative probabilities. Put a flag mode f in A and consider

$$|\Psi\rangle = \sqrt{1-\epsilon} |0\rangle_f |a_0\rangle_{A'} |b_0\rangle_{\bar{A}} + \sqrt{\epsilon} |1\rangle_f |\Phi_L\rangle_{A'\bar{A}}, \quad A = f \cup A'. \quad (6)$$

Choose $S_1(\Phi_L) = \Theta(L)$ and the $\bar{A}$ Schmidt support of Φ_L orthogonal to $|b_0\rangle$. The reduced eigenvalues are $1-\epsilon$ and ϵ times those of $\rho_{A'}^{\Phi_L}$, hence

$$S_1(A; \Psi) = H_2(\epsilon) + \epsilon S_1(A'; \Phi_L), \quad H_2(\epsilon) = -\epsilon \log \epsilon - (1-\epsilon) \log(1-\epsilon). \quad (7)$$

For $\epsilon = O(L^{-1})$, the input entropy is $O(1)$. Measuring the flag and obtaining 1 leaves entropy $\Theta(L)$, even though the operation is entirely in A . The two components can be chosen in the same total fermion-parity sector, so the example needs no parity-violating measurement.

The initial Schmidt rank was already $1 + \chi_A(\Phi_L)$. There is no increase in S_0 . Thus a branchwise bound on ΔS_0 is not a bound on ΔS_1 . For a product input, however, $S_0^{\text{in}} = 0$, and bounding S_0^{out} bounds every S_q^{out} .

II. CAPACITY OF AN INDIVIDUAL CONDITIONAL OPERATION

Across the same bipartition, expand an operation as

$$\hat{O} = \sum_{\alpha=1}^r \sigma_\alpha \hat{O}_\alpha^A \otimes \hat{O}_\alpha^{\bar{A}}. \quad (8)$$

The smallest number of terms is the operator-Schmidt rank r . Equivalently, reshape the matrix elements so that the input/output indices on A form one index and those on $\bar{A}$ form the other; the rank of this reshaped matrix is r . Any explicit product expansion supplies an upper bound.

Applying Eq. (8) to Eq. (4) gives

$$\hat{O}|\psi\rangle = \sum_{\alpha=1}^r \sum_{i=1}^{\chi_A} \sigma_\alpha \sqrt{p_i} (\hat{O}_\alpha^A |\phi_i\rangle_A) \otimes (\hat{O}_\alpha^{\bar{A}} |\theta_i\rangle_{\bar{A}}). \quad (9)$$

The output is a sum of at most $r\chi_A$ product vectors. They need not be orthogonal or independent, so its Schmidt rank is at most this number. Normalization multiplies the coefficient matrix by a scalar. Therefore

$$\chi_A^{\text{out}} \leq r\chi_A^{\text{in}}, \quad S_0^{\text{out}} - S_0^{\text{in}} \leq \log r. \quad (10)$$



FIG. 1. (a) Only cut-crossing operations can increase Schmidt rank; a crossing operation of operator-Schmidt rank r_j multiplies it by at most r_j . (b) A full-width strip on a torus has two cuts of length N_x . Localization near the walls does not reduce the subsystem boundary to two points. Opposite edges of the rectangle are periodically identified.

No inverse is used: singular projectors are allowed. Zero-norm branches are excluded only because they cannot be normalized.

An operation supported on one side has a one-term product representation. For an odd fermionic operator on $\bar{A}$, its representation contains a parity factor on A ; it is still one product term. The relevant support is physical fermionic support, not the apparent length of a parity string.

Let $\mathcal{C}_\partial(\mathbf{m})$ contain the operations whose physical supports cross the cut on a realized record. Iteration gives

$$\chi_A(\psi_{\mathbf{m}}) \leq \chi_A(\psi_{\text{in}}) \prod_{j \in \mathcal{C}_\partial(\mathbf{m})} r_j, \quad (11)$$

$$S_0(A; \psi_{\mathbf{m}}) - S_0(A; \psi_{\text{in}}) \leq \sum_{j \in \mathcal{C}_\partial(\mathbf{m})} \log r_j. \quad (12)$$

Feedback changes the realized sequence but not the proof for that sequence. This bound needs no minimum outcome probability, Gaussianity, translation invariance, or invertibility.

III. FROM CROSSING OPERATIONS TO PHYSICAL CIRCUIT DEPTH

Suppose each operation has support diameter at most R , onsite Hilbert spaces have dimension d_{loc} , and the lattice has bounded coordination. Fix R and d_{loc} as the system grows. A physical layer contains mutually nonoverlapping quantum operations. A crossing gate meeting s_A sites in A and $s_{\bar{A}}$ in $\bar{A}$ has

$$r_j \leq \min\{d_{\text{loc}}^{2s_A}, d_{\text{loc}}^{2s_{\bar{A}}}\} \leq r_*, \quad (13)$$

because these are its two local operator-space dimensions. For fixed range, r_* is size independent.

Crossing gates meet the boundary neighborhood of thickness R , whose volume on a regular lattice is bounded by a constant times $|\partial A|$. Nonoverlap limits their number per layer to $\nu_\partial |\partial A|$, where ν_∂ is size independent. For D_{circ} layers,

$$\Delta S_0(A) \leq D_{\text{circ}} \nu_\partial |\partial A| \log r_* = \kappa D_{\text{circ}} |\partial A|. \quad (14)$$

This is the entanglement-growth bound of Ref. [1], Appendix D. For pure product inputs it implies

$$S_q^{\text{out}}(A) \leq \kappa D_{\text{circ}} |\partial A|, \quad q > 0. \quad (15)$$

Local product ancillas add no initial cross-cut rank. A fixed ancillary overhead is absorbed in d_{loc} and κ . If an entangled ancilla is discarded without a resolved outcome, the physical output can be mixed: one must specify a purification and its cut before using the pure-state argument. The controller below instead leaves its outgoing ancilla factorized on every record.

Physical depth is not a simulation update count. Overlapping OW measurements in one sweep cannot all count as one quantum layer. Bounded-support patches admit a finite coloring, but reordering noncommuting measurements can change the protocol; a coloring alone does not establish the depth of the original ordered sweep. Classical communication can be nonlocal in this bound. Including its physical latency adds a time cost without weakening the rank inequality.

IV. OCCUPATION MEASUREMENTS WITH FERMIONIC-SWAP CORRECTION

A. The four conditional operators

Consider one normalized overcomplete Wannier (OW) mode in BPJ's fill-lower-band, deplete-upper-band construction [2]. Suppressing its position, family, and band labels, write

$$\hat{\chi} = \sum_{x \in X} \chi_x \hat{c}_x, \quad \sum_{x \in X} |\chi_x|^2 = 1, \quad \{\hat{c}_x^\dagger, \hat{c}_y\} = \delta_{xy}. \quad (16)$$

The index x includes orbitals. The patch X is obtained by finite-range truncation and renormalization. Distinct OW modes are generally nonorthogonal, so their number operators need not commute. Only the single-mode algebra is used in the following step. Define

$$\hat{N}_\chi = \hat{\chi}^\dagger \hat{\chi}, \quad \hat{\Pi}_1 = \hat{N}_\chi, \quad \hat{\Pi}_0 = \hat{\mathbf{1}} - \hat{N}_\chi. \quad (17)$$

The relations $\hat{\chi}^2 = 0$ and $\hat{\chi} \hat{\chi}^\dagger = \hat{\mathbf{1}} - \hat{N}_\chi$ imply $\hat{N}_\chi^2 = \hat{N}_\chi$, so these are complementary projectors.

The target is $s = 1$ for filling and $s = 0$ for depletion. Measure m and do nothing when $m = s$. For a mismatch, supply a fresh local ancillary mode $\hat{d}$ in $|s\rangle_a$ and apply fSWAP($\hat{\chi}, \hat{d}$). In the ordered two-mode Fock basis,

$$|0, 0\rangle \mapsto |0, 0\rangle, \quad |1, 0\rangle \mapsto |0, 1\rangle, \quad |0, 1\rangle \mapsto |1, 0\rangle, \quad |1, 1\rangle \mapsto -|1, 1\rangle. \quad (18)$$

The last minus sign is fermionic exchange. A mismatch has one particle in this sector, hence $|m, s\rangle \mapsto |s, m\rangle$. Its outgoing ancilla has definite occupation and factorizes from the physical state.

Taking the graded ancillary matrix element with a consistent branch-phase choice yields

$$\begin{array}{c|cc} & m=0 & m=1 \\ \hline s=0 & \hat{K}_{0|0} = \hat{\mathbf{1}} - \hat{N}_\chi & \hat{K}_{1|0} = \hat{\chi} \\ s=1 & \hat{K}_{0|1} = \hat{\chi}^\dagger & \hat{K}_{1|1} = \hat{N}_\chi \end{array}. \quad (19)$$

For depletion, $\hat{\chi} \hat{\Pi}_1 = \hat{\chi}$; for filling, $\hat{\chi}^\dagger \hat{\Pi}_0 = \hat{\chi}^\dagger$. The joint fSWAP conserves charge and parity. The reduced mismatch operators are odd because the ancilla supplies or removes a particle. In an ordinary tensor-product encoding, ancillary matrix elements may contain an input-parity factor; on a definite-parity branch it is an irrelevant scalar phase.

Directly using the single-mode algebra gives

$$\hat{K}_{m|s}^\dagger \hat{K}_{m|s} = \hat{\Pi}_m, \quad \sum_m \hat{K}_{m|s}^\dagger \hat{K}_{m|s} = \hat{\mathbf{1}}, \quad \hat{\Pi}_s \hat{K}_{m|s} = \hat{K}_{m|s}. \quad (20)$$

Thus the original Born probabilities are preserved and every nonzero branch attains its target occupation. Perfect correction does not make the measurement outcomes deterministic.

This fresh-ancilla model differs from BPJ's finite ancillary bath. There, ancillary occupation redistribution and measurements determine whether a suitable particle or hole is available; Appendix D of Ref. [2] develops a Markovian approximation to that bath. Here the required occupation is supplied separately for every correction, and the outgoing factorized ancilla can be discarded.

B. Parity strings and cut-crossing ranks

Order all A modes before $\bar{A}$ modes, with $\hat{P}_A = (-1)^{\hat{N}_{F,A}}$. Splitting the OW coefficients gives

$$\hat{\chi} = \hat{a}_A \otimes \hat{\mathbf{1}}_{\bar{A}} + \hat{P}_A \otimes \hat{a}_{\bar{A}}. \quad (21)$$

The partial modes are not normalized separately: $\{\hat{a}_A, \hat{a}_A^\dagger\} = \sum_{x \in A} |\chi_x|^2$ and similarly on $\bar{A}$. The parity factor anticommutes with $\hat{a}_A$ and enforces the anticommutation of physical fermions on opposite sides. There are two product terms, so

$$r(\hat{\chi}) \leq 2, \quad r(\hat{\chi}^\dagger) \leq 2. \quad (22)$$

Multiplying Eq. (21) by its adjoint in the appropriate order, without moving parity factors past odd operators, gives

$$\hat{N}_\chi = \hat{a}_A^\dagger \hat{a}_A \otimes \hat{\mathbf{1}}_{\bar{A}} + \hat{a}_A^\dagger \hat{P}_A \otimes \hat{a}_{\bar{A}} + \hat{P}_A \hat{a}_A \otimes \hat{a}_{\bar{A}}^\dagger + \hat{\mathbf{1}}_A \otimes \hat{a}_{\bar{A}}^\dagger \hat{a}_{\bar{A}}. \quad (23)$$

Subtracting from the identity needs no fifth term:

$$\hat{\mathbf{1}} - \hat{N}_\chi = (\hat{\mathbf{1}}_A - \hat{a}_A^\dagger \hat{a}_A) \otimes \hat{\mathbf{1}}_{\bar{A}} - \hat{a}_A^\dagger \hat{P}_A \otimes \hat{a}_{\bar{A}} - \hat{P}_A \hat{a}_A \otimes \hat{a}_{\bar{A}}^\dagger - \hat{\mathbf{1}}_A \otimes \hat{a}_{\bar{A}}^\dagger \hat{a}_{\bar{A}}. \quad (24)$$

We therefore obtain

$$r(\hat{N}_\chi) \leq 4, \quad r(\hat{\mathbf{1}} - \hat{N}_\chi) \leq 4. \quad (25)$$

These are upper bounds, not necessarily the minimal ranks for special orbitals. If the mode lies on one side, every branch has rank one.

C. Record-resolved capacity

Let $N_{=}^\partial(\mathbf{m})$ count crossing events with $m = s$ and $N_{\neq}^\partial(\mathbf{m})$ count those with $m \neq s$. For a circuit consisting of the corrected measurements above, Eq. (12) becomes

$$\Delta S_0^m(A) \leq (\log 2) [2N_{=}^\partial(\mathbf{m}) + N_{\neq}^\partial(\mathbf{m})]. \quad (26)$$

The projector branch has the larger upper bound even though it requires no feedback: measuring a mode spread across the cut can itself entangle the sides. Neither rank bound is an entropy necessarily generated by a gate. Extra crossing operations, if present, require their own $\log r_j$ terms.

V. APPLICATION TO CRITICAL INTERFACES IN TWO DIMENSIONS

A. Geometry and the conformal assumption

On an $N_x \times N_y$ torus with two walls running along y , the full-width strip and its boundary are

$$A_\ell = \{(x, y) : 0 \leq x < N_x, 0 \leq y < \ell\}, \quad |\partial A_\ell| = 2N_x. \quad (27)$$

Its two cuts are transverse to the walls [Fig. 1(b)]. Even when an entanglement contour concentrates near the walls, the circuit is local in the two-dimensional lattice. This geometry does not become an intrinsically one-dimensional preparation problem.

Make the additional infrared assumption that the prepared pure state has the vacuum-like interval scaling of independent conformal branches. Let c_Σ sum the left- and right-moving central charges over all walls intersected by the strip. Then [3, 4]

$$S_1(A_\ell) = b(N_x) + \frac{c_\Sigma}{6} \log \left[\frac{N_y}{\pi a} \sin \left(\frac{\pi \ell}{N_y} \right) \right] + o(1). \quad (28)$$

Here a is a microscopic cutoff and $b(N_x)$ includes short-distance boundary contributions. This is not a consequence of Gaussianity or of the rank theorem. It presumes a scale window where lattice corrections, interwall coupling, and finite correlation length do not destroy the conformal form. At fixed width an eventual interwall gap can end that window; an asymptotic claim requires scaling to persist.

For oppositely oriented charge-one complex-fermion walls, $c_\Sigma = 2$ even though their signed chiral central charges cancel. Their positive entropies add, giving the full-strip coefficient $1/3$.

B. Necessary depth and crossing-event resources

At half circumference and fixed N_x , $S_1 = (c_\Sigma/6) \log N_y + O(1)$. Combining this with the product-input bound gives

$$2\kappa D_{\text{circ}} N_x \geq \frac{c_\Sigma}{6} \log N_y - O(1), \quad D_{\text{circ}} N_x = \Omega(\log N_y). \quad (29)$$

If N_x also grows, one needs a uniform positive logarithmic lower bound: a width-dependent offset cannot silently be treated as $O(1)$. A sufficient assumption is $S_1(A_{N_y/2}) \geq (c_\Sigma/6) \log N_y - C$ with size-independent C along the chosen family.

At fixed width the required physical depth is at least logarithmic. If N_x itself supplies boundary capacity of order $\log N_y$, the bound no longer rules out constant depth. It neither constructs such a circuit nor shows that the present controller attains it.

Using Eq. (26) instead gives, on each trajectory with the stated entropy,

$$2N_{=}^{\partial}(\mathbf{m}) + N_{\neq}^{\partial}(\mathbf{m}) \geq \frac{c_\Sigma}{6 \log 2} \log N_y - O(1). \quad (30)$$

The coefficient is $1/(3 \log 2)$ for two complex-fermion walls. If only the Born-averaged entropy has the logarithmic scaling, average the branchwise upper bound:

$$\overline{S_1(A_\ell)} \leq (\log 2) \left(2\overline{N_{=}^{\partial}} + \overline{N_{\neq}^{\partial}} \right). \quad (31)$$

This constrains the mean resource, not the entropy of every record. It is also not a statement about the entropy of the discarded-record mixture.

C. Initial entanglement, mixed inputs, and finite range

For a general pure input, the correct statement is

$$S_q^{\text{out}}(A) \leq S_0^{\text{in}}(A) + \sum_{j \in \mathcal{C}_\partial} \log r_j. \quad (32)$$

A random Slater determinant is already spatially entangled and can have large Schmidt rank. It need not satisfy a product-preparation lower bound. For a mixed input, attach a reference and specify its assignment to the cut; the theorem concerns that enlarged pure state. The physical subsystem entropy alone is not a mixed-state entanglement measure.

Fixed-range truncation is essential to size-independent constants in Eq. (14). Exponentially decaying but nonzero OW tails are not bounded support, and an arbitrarily small tail can change exact Schmidt rank. A truncation-error estimate for ordinary observables does not by itself control S_0 . Growing range, finite-accuracy preparation, and the locality overhead of a fermion-to-qubit encoding require separate analysis.

These results identify a necessary entangling resource. They do not prove that the wall is conformal, determine a convergence rate, or equate physical depth with the number of simulated cycles.

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